

Text-Guided Video Fragment Localization with Adapted CLIP Embeddings

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Abstract

This project studies text-guided video fragment localization: given a full video and a natural-language caption, the goal is to predict the temporal interval where the described event occurs. A pretrained X-CLIP model is used as the baseline video-language encoder and is kept frozen during training. Smaller trainable reranking models are then trained on top of X-CLIP video and text embeddings to score candidate time segments from ActivityNet Captions. Three systems are compared under identical validation conditions: the pretrained X-CLIP baseline, a supervised Segment Reranker, and a Transformer-based Contextual Reranker. On a validation set of 400 ActivityNet videos containing 1,272 captioned moments, raw X-CLIP achieved a mean IoU of 0.214 when only the highest-ranked prediction was evaluated, and 0.396 when the best of five returned candidates was considered. The Segment Reranker improved these scores to 0.354 and 0.551, while the Contextual Reranker reached 0.361 and 0.561. The results show that X-CLIP embeddings help retrieve relevant regions, but exact single-interval prediction remains difficult.

1 Introduction

Video fragment localization is the problem of finding when a natural-language event description occurs inside an untrimmed video. Unlike video retrieval, which returns relevant videos for a query, this task must return the start and end time of the matching event inside a given video. It is important for video search, editing, summarization, dataset annotation, and assistive browsing of long video collections.

The task is difficult because language descriptions can refer to short actions, long activities, scene changes, repeated motions, or outcomes that are only visible after several steps. A video may contain multiple visually similar fragments, and the correct answer depends on temporal boundaries, not just semantic similarity. ActivityNet Captions further increases the challenge because captions from the same video often describe overlapping or neighboring events at different temporal scales.

This project uses a pretrained X-CLIP model as a frozen feature extractor, then trains smaller models on top of the X-CLIP embeddings to rank candidate video segments. The first implementation used image CLIP [11]: frames were sampled from videos, encoded independently, and compared with the query text. This simple frame-level baseline was weak because single images often miss motion, temporal context, and events that unfold over several seconds. The pipeline was therefore changed to X-CLIP [10], which encodes short video clips rather than isolated frames.

X-CLIP maps video clips and text into a shared embedding space, but it is designed mainly for video-text retrieval, not direct temporal

grounding. The contribution of this work is a pipeline that uses X-CLIP as a fixed feature extractor, creates possible video segments at several durations, and trains smaller models to rank the segment that best matches the caption.

2 Related Work

CLIP[11] demonstrated that contrastive image-text pretraining can produce embeddings that are useful for many downstream tasks. X-CLIP [10] extends this idea to video-text retrieval by learning shared representations for videos and text, while CLIP4Clip [7] studies CLIP adaptation for video retrieval. These models are effective for matching videos with relevant captions, but retrieval is not the same as localization: a model can recognize that a caption matches a video and still fail to identify the exact temporal interval.

ActivityNet [2] and ActivityNet Captions [5] are standard benchmarks for untrimmed video understanding with localized language annotations. Early temporal grounding methods such as TALL/CTRL [4] and natural-language moment localization [1] combine video-language matching with temporal proposal scoring. Later work improves temporal reasoning through richer proposal or boundary representations. 2D-TAN [14] models start/end combinations as a two-dimensional temporal map. LGI [9] uses local-global video-text interactions. DRN [12] performs dense regression for video grounding, and VSLNet [13] treats localization as span prediction over video features.

Recent DETR-style approaches make moment prediction more direct. DETR [3] introduced Transformer-based set prediction for object detection, and later video models adapted this idea to predict time intervals for text queries. Moment-DETR [6] and QD-DETR [8] use specialized Transformer-based architectures to predict video time intervals for text queries. Those systems are closer to state of the art, but typically require prepared feature files, task-specific model code, and a larger training setup.

3 Experiments

3.1 Dataset

Experiments use ActivityNet Captions [5]. The training set contains 2,000 ActivityNet training videos with 6,807 captioned moments. The validation set was created from 400 ActivityNet validation samples and contains 1,272 captioned moments. Each example contains a video path, caption, ground-truth start time, and ground-truth end time.

3.2 Model Architectures

All three systems use the same frozen X-CLIP model `microsoft/xclip-base-patch32`. Video windows are sampled at 4, 8, 16, and

32 seconds. The step size between consecutive windows is 1, 2, 4, and 8 seconds respectively. Each video window is encoded with X-CLIP, and each caption is encoded into the same 512-dimensional embedding space.

Raw X-CLIP is the pretrained baseline. It scores temporal windows using video-text similarity and merges high-scoring windows into temporal segments. It does not use any training on the ActivityNet timestamp labels.

Segment Reranker is an MLP-based scoring model trained using the ActivityNet ground-truth timestamps. Candidate video segments are created by grouping neighboring X-CLIP windows. The allowed group sizes are $\{1, 2, 3, 4, 6, 8, 12, 16, 24, 32\}$ windows, so the model can evaluate both short and long candidate intervals. Each candidate segment is represented by the average X-CLIP embedding of the windows inside it, together with scalar features such as duration, relative start time, relative end time, center position, and X-CLIP similarity statistics.

The architecture has two projection branches. One branch maps the 512-dimensional video segment embedding to 256 dimensions, and the other maps the 512-dimensional caption embedding to 256 dimensions. Each branch consists of a linear layer, layer normalization, and GELU activation. The projected video embedding, projected caption embedding, their absolute difference, their elementwise product, and the scalar features are concatenated. This combined vector is passed through a score head with a linear layer, GELU activation, dropout, and a final linear layer that outputs one match score for the candidate segment.

Contextual Reranker extends the Segment Reranker by adding a temporal Transformer encoder before candidate scoring. Instead of scoring each candidate only from its averaged X-CLIP segment embedding, it first builds context-aware representations for all windows in the video.

The architecture begins by projecting each 512-dimensional X-CLIP video window embedding and the 512-dimensional caption embedding to 256 dimensions. For each window, the model combines the projected video embedding, projected caption embedding, their absolute difference, their elementwise product, and temporal position features. This representation is passed through a two-layer Transformer encoder with four attention heads. Candidate segments are then represented by average pooling the contextualized window embeddings inside the candidate interval. Finally, this pooled segment representation is concatenated with scalar segment features and the base Segment Reranker score, and an MLP score head predicts one match score for each candidate segment.

3.3 Training and Evaluation

X-CLIP remains frozen in all experiments. Embeddings are cached to disk for efficiency, so training updates only the smaller scoring models. The Segment Reranker is trained with IoU-based supervision: high-overlap candidates are positives, and low-overlap or confusing candidates are negatives. The Contextual Reranker is trained after the Segment Reranker and uses the Segment Reranker as a base scorer.

Performance is measured with temporal Intersection over Union (IoU), as defined in Appendix A. The main localization result is the highest-ranked prediction, because that corresponds to a single

final timestamp interval. Results are also reported when the best interval among several returned candidates is considered. This retrieval-style metric is common in moment localization and helps diagnose whether a method fails because it cannot generate a good candidate or because it ranks a good candidate below worse ones.

Table 1: Validation results on 1,272 ActivityNet Captions video intervals.

Method	Top- k	mIoU	R@0.1	R@0.5	R@0.7
Raw X-CLIP	1	0.214	0.494	0.171	0.075
Raw X-CLIP	2	0.303	0.667	0.260	0.114
Raw X-CLIP	3	0.353	0.760	0.314	0.141
Raw X-CLIP	4	0.380	0.809	0.342	0.151
Raw X-CLIP	5	0.396	0.829	0.367	0.161
Segment Reranker	1	0.354	0.743	0.314	0.149
Segment Reranker	2	0.446	0.816	0.457	0.245
Segment Reranker	3	0.494	0.848	0.537	0.305
Segment Reranker	4	0.527	0.872	0.593	0.347
Segment Reranker	5	0.551	0.888	0.631	0.386
Contextual Reranker	1	0.361	0.745	0.324	0.156
Contextual Reranker	2	0.450	0.825	0.454	0.242
Contextual Reranker	3	0.503	0.861	0.545	0.314
Contextual Reranker	4	0.540	0.883	0.613	0.366
Contextual Reranker	5	0.561	0.902	0.649	0.390

4 Discussion

The learned models clearly improve over raw X-CLIP. At Top-1, mean IoU rises from 0.214 for raw X-CLIP to 0.354 for the Segment Reranker and 0.361 for the Contextual Reranker. At Top-5, mean IoU rises from 0.396 to 0.551 and 0.561 respectively. The same pattern appears in recall: Contextual Reranker Top-5 reaches 0.649 recall at IoU 0.5, compared with 0.367 for raw X-CLIP. For context, 2D-TAN reports 44.51% R@1 and 77.13% R@5 at IoU 0.5 on ActivityNet Captions, while the Contextual Reranker reaches 32.39% and 64.94% under the same IoU threshold [14].

The results also show the central limitation. The best of five scores are much stronger than the single-prediction scores, which means good candidates are often present but not always ranked first. This justifies reporting results for several returned candidates: it shows that the pipeline can often find a reasonable region, but its scoring model is not reliable enough to always choose the correct region as the final answer. The remaining problem is therefore precise ranking and boundary selection. The Segment Reranker learns dataset-specific temporal scoring, and the Contextual Reranker improves it by using temporal context, but frozen X-CLIP embeddings still lack the boundary precision of dedicated temporal grounding models.

Overall, the project demonstrates that adapted X-CLIP embeddings can produce a useful temporal retrieval system with modest training cost. However, reaching substantially higher single-prediction IoU would likely require changing the baseline architecture more fundamentally. Future work should explore models designed to predict time intervals directly, such as Moment-DETR or QD-DETR, use stronger video features, or fine-tune the full video-language model instead of keeping X-CLIP fixed.

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A Temporal IoU

For a predicted interval $P = [s_p, e_p]$ and a ground-truth interval $G = [s_g, e_g]$, temporal Intersection over Union (IoU) is defined as

$$\text{IoU}(P, G) = \frac{\max(0, \min(e_p, e_g) - \max(s_p, s_g))}{\max(e_p, e_g) - \min(s_p, s_g)}.$$

This value is 1 when the predicted and ground-truth intervals match perfectly, and 0 when they do not overlap.